Email Server Configuration using Cisco Packet Tracer

Submitted By

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Aim

To design and configure a small network using router, switch, server, and two PCs in Cisco Packet Tracer, create two different IP networks, configure routing between them, set up an Email Server using SMTP and POP3, create user accounts, and verify email communication between two PCs.

Requirements

Devices

- 1 Router (e.g., 2911)
- 1 Switch (e.g., 2960)
- 1 Server
- 2 PCs (PC0 and PC1)

Software

- Cisco Packet Tracer

Cables

- Copper Straight-Through cables

Procedure

1. Network Topology Creation and Physical Connections

1. Open Cisco Packet Tracer.
2. Add the following devices:
   - Router, Switch, Server, PC0, PC1
3. Connect the devices using Copper Straight-Through cables:

- PC0 → Switch
- PC1 → Switch
- Switch → Router
- Router → Server

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2. IP Addressing Scheme

Device	Interface	IP Address	Subnet Mask	Default Gateway
PC0	FastEthernet0	192.168.1.2	255.255.255.0	192.168.1.1
PC1	FastEthernet0	192.168.1.3	255.255.255.0	192.168.1.1
Router	G0/0	192.168.1.1	255.255.255.0	-
Router	G0/1	192.168.2.1	255.255.255.0	-
Server	FastEthernet0	192.168.2.2	255.255.255.0	192.168.2.1

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3. Configure PC IP Addresses (Static)

1. Click PC0 → Desktop → IP Configuration.
2. Enter:
   - IP Address: 192.168.1.2
   - Subnet Mask: 255.255.255.0
   - Default Gateway: 192.168.1.1
3. Repeat for PC1:
   - IP Address: 192.168.1.3
   - Subnet Mask: 255.255.255.0
   - Default Gateway: 192.168.1.1

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4. Configure Server IP Address

1. Click Server → Desktop → IP Configuration.
2. Enter:
   o IP Address: 192.168.2.2
   o Subnet Mask: 255.255.255.0
   o Default Gateway: 192.168.2.1

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5. Configure Router Interfaces

1. Click Router → CLI.
2. Enter the following commands:

```shell
enable
configure terminal

interface g0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit

interface g0/1
ip address 192.168.2.1 255.255.255.0
no shutdown
exit

end
```

This enables communication between the two networks i.e. the PC Network and the Server Network.

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6. Configure Email Server

1. Click Server → Services → Email
2. Turn SMTP = ON.
3. Turn POP3 = ON.
4. Set Domain Name (example): `iiitk.com`

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7. Create User Accounts

Under Email Server Users, create:

Username	Password
user0	123
user1	123

Email IDs:

- user0@iiitk.com
- user1@iiitk.com

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8. Configure Email Client on PCs

1. Go to **PC0/1 → Desktop → Email**.
2. Configure:

PC0

Field	Value
Your Name	User0
Email Address	user0@iiitk.com
Incoming Mail Server	192.168.2.2
Outgoing Mail Server	192.168.2.2
Username	user0
Password	123

PC1

Field	Value
Your Name	User1
Email Address	user1@iiitk.com
Incoming Mail Server	192.168.2.2
Outgoing Mail Server	192.168.2.2
Username	user1
Password	123

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9. Test Email Communication

Send Email from PC0 to PC1

1. Open Email → Compose on PC0.
2. Enter:
   - To: user1@iiitk.com
   - Subject: Test Mail
   - Message: Hello from PC0
3. Click Send.

Reply from PC1 to PC0

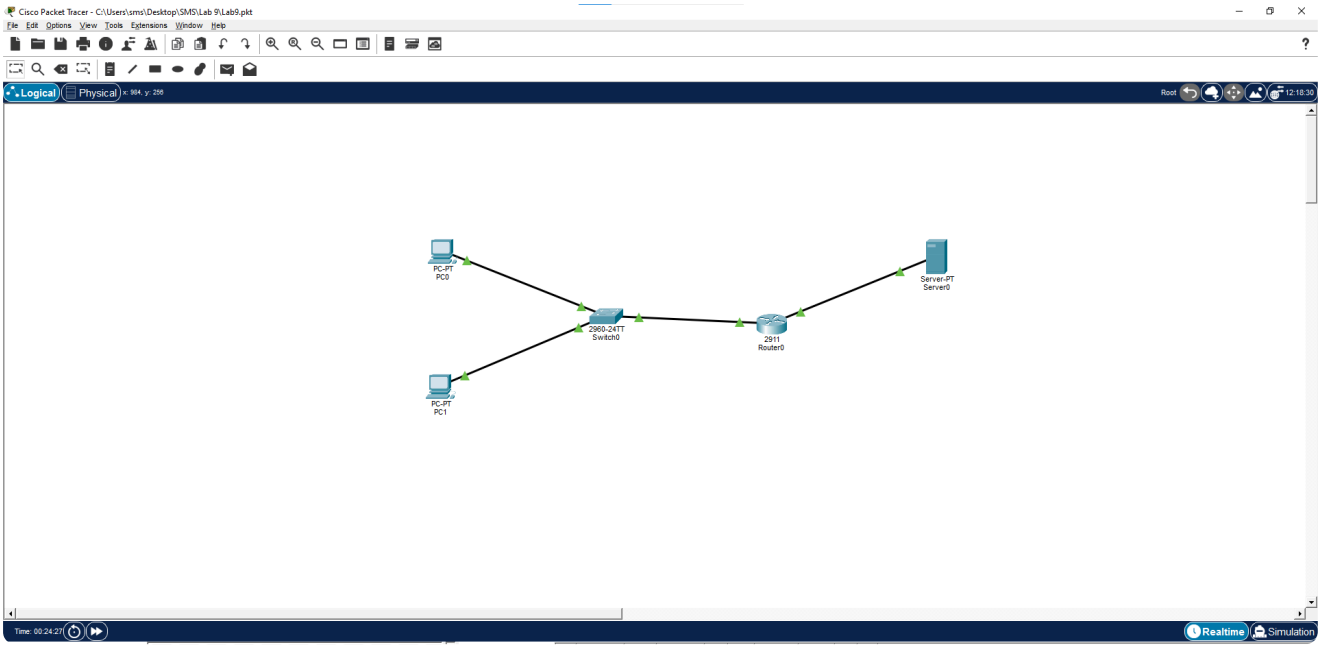
1. Open PC1 → Email → Receive.
2. Open the received mail.
3. Click Reply.

4. Type message:
   o Reply from PC1
5. Click Send.

Output Screenshots

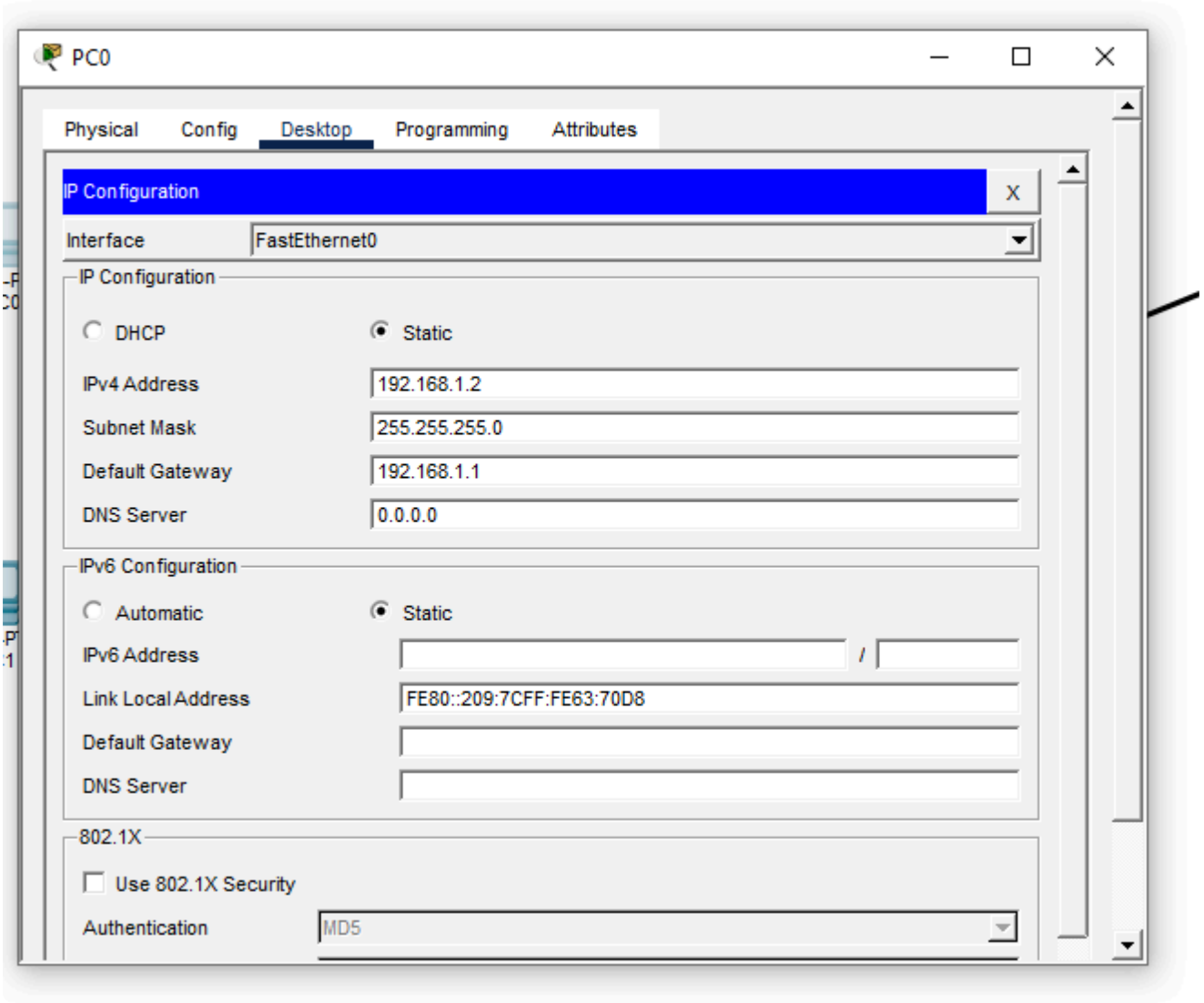
1. Network Topology

Router, Switch, Server, PC0, PC1 connected.

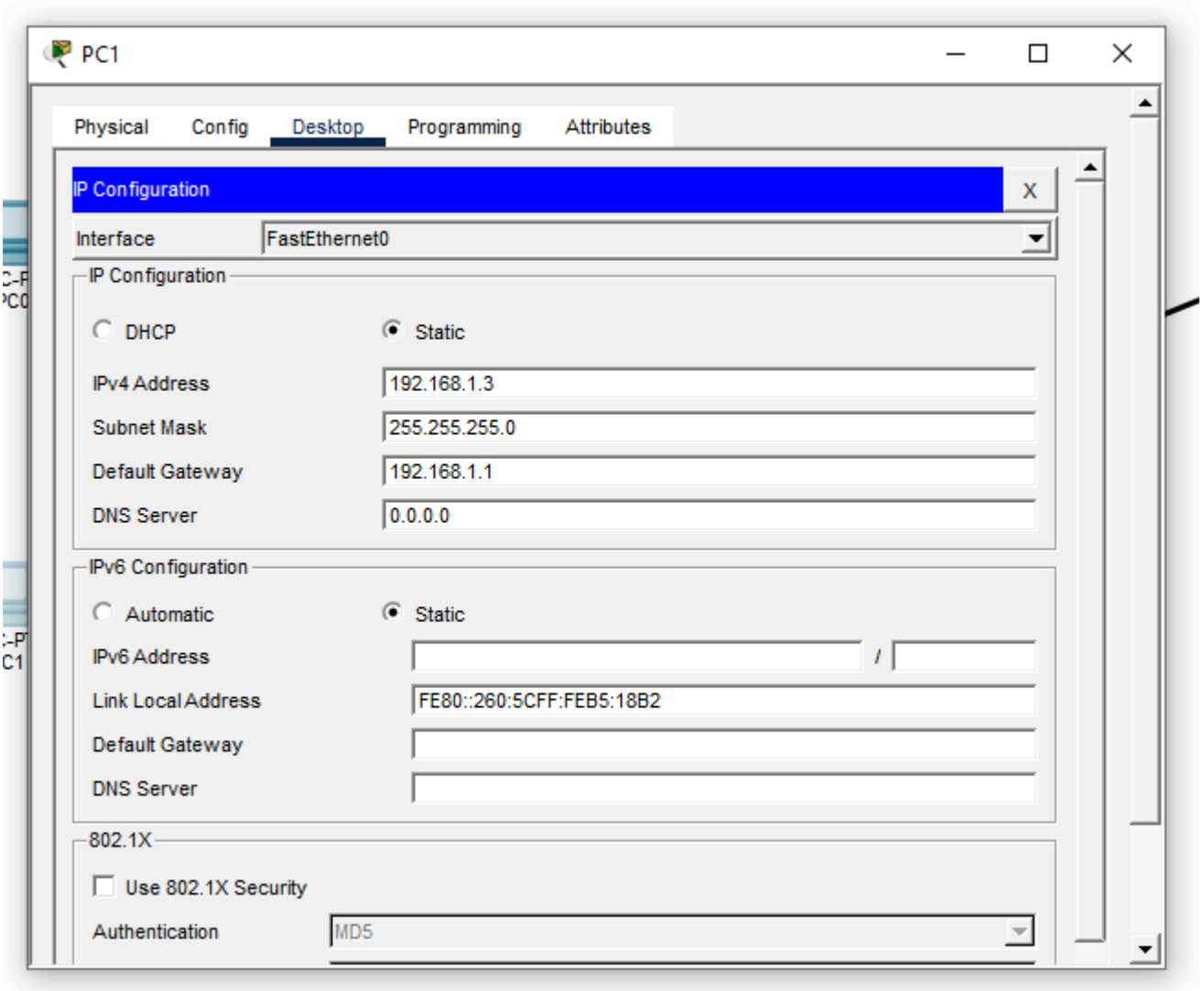


2. IP Configuration

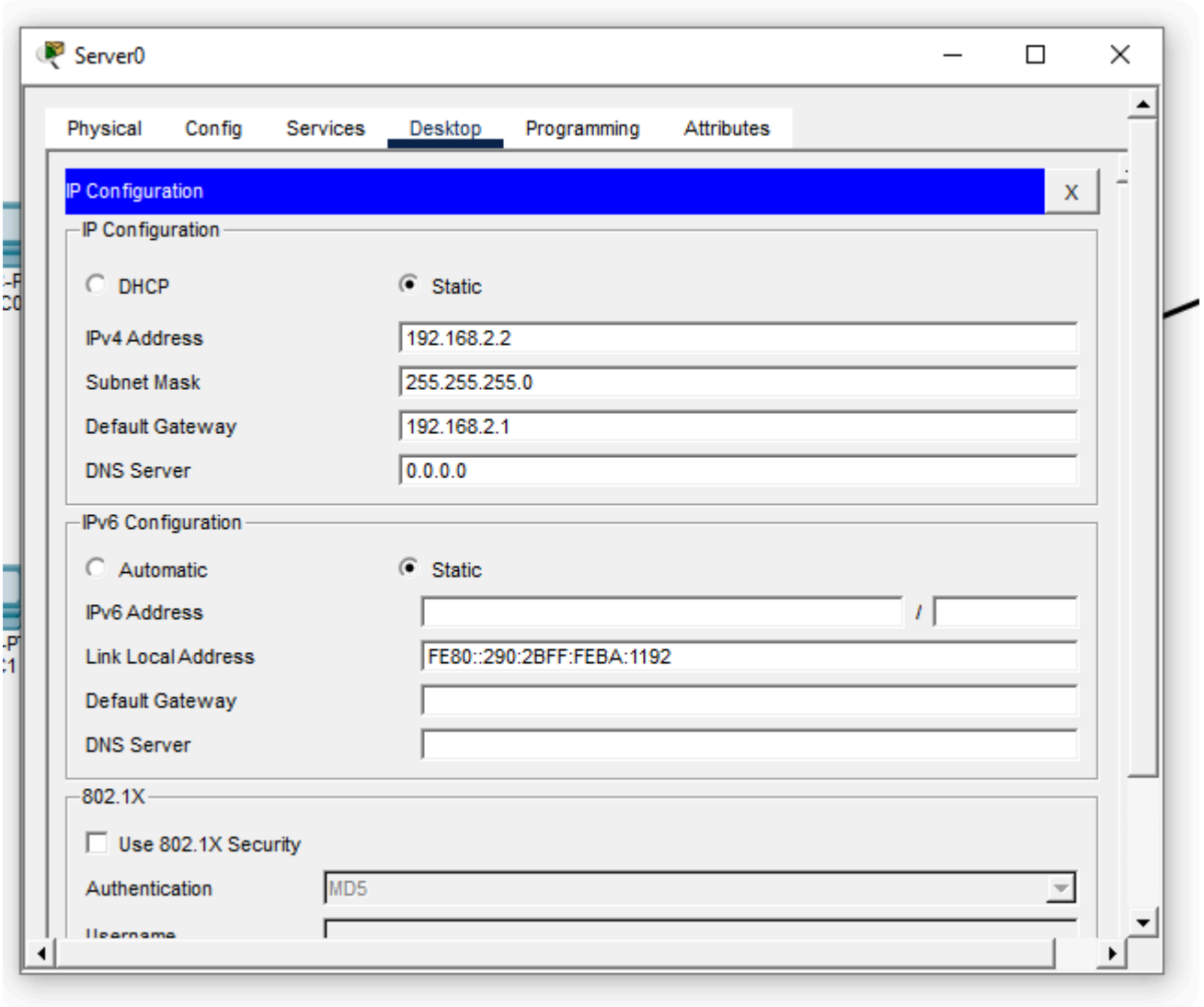
PC0:


PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.1.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::209:7CFF:FE63:70D8
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5

PC1:

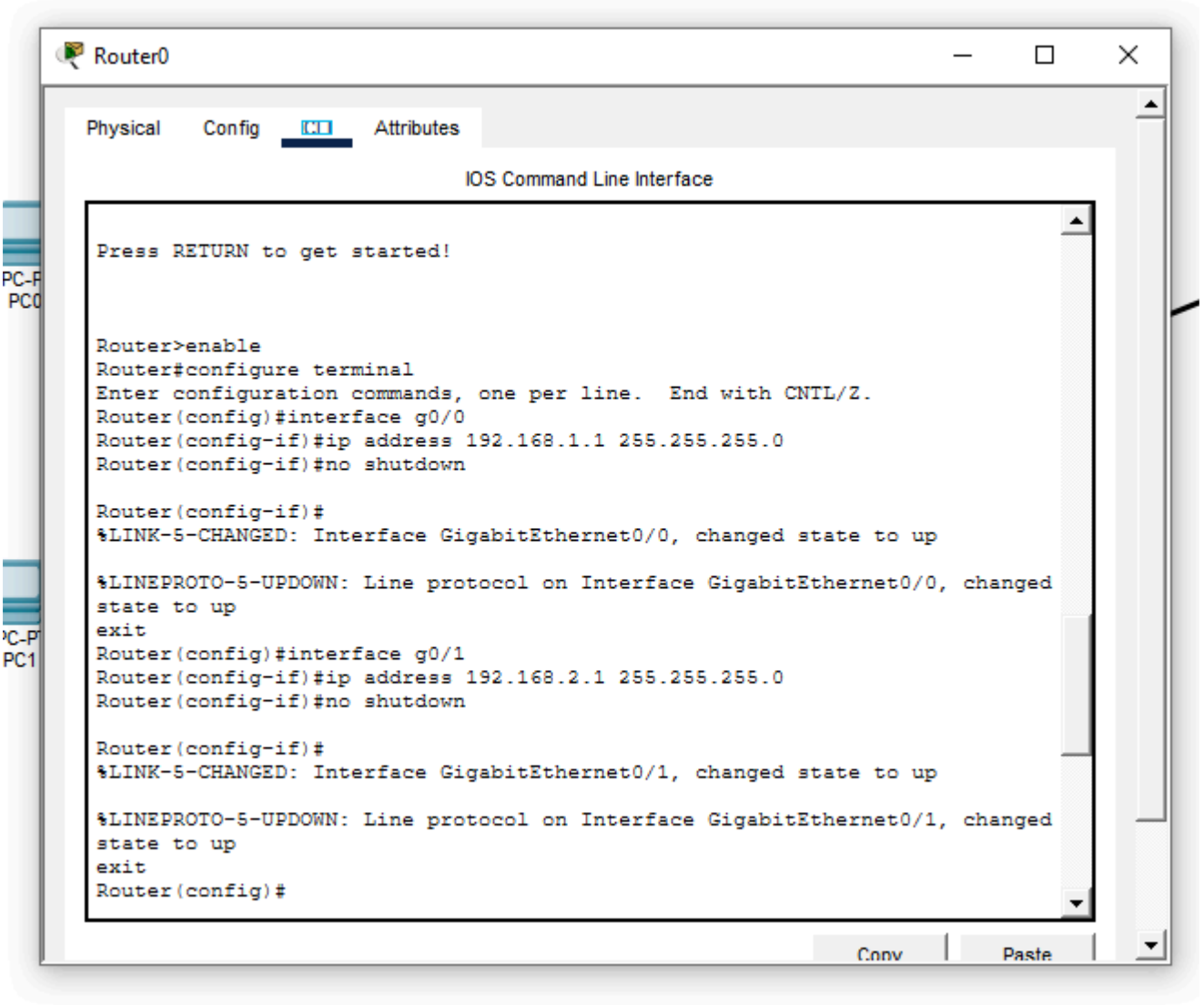

PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.3
Subnet Mask
255.255.255.0
Default Gateway
192.168.1.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::260:5CFF:FEB5:18B2
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5

Server:



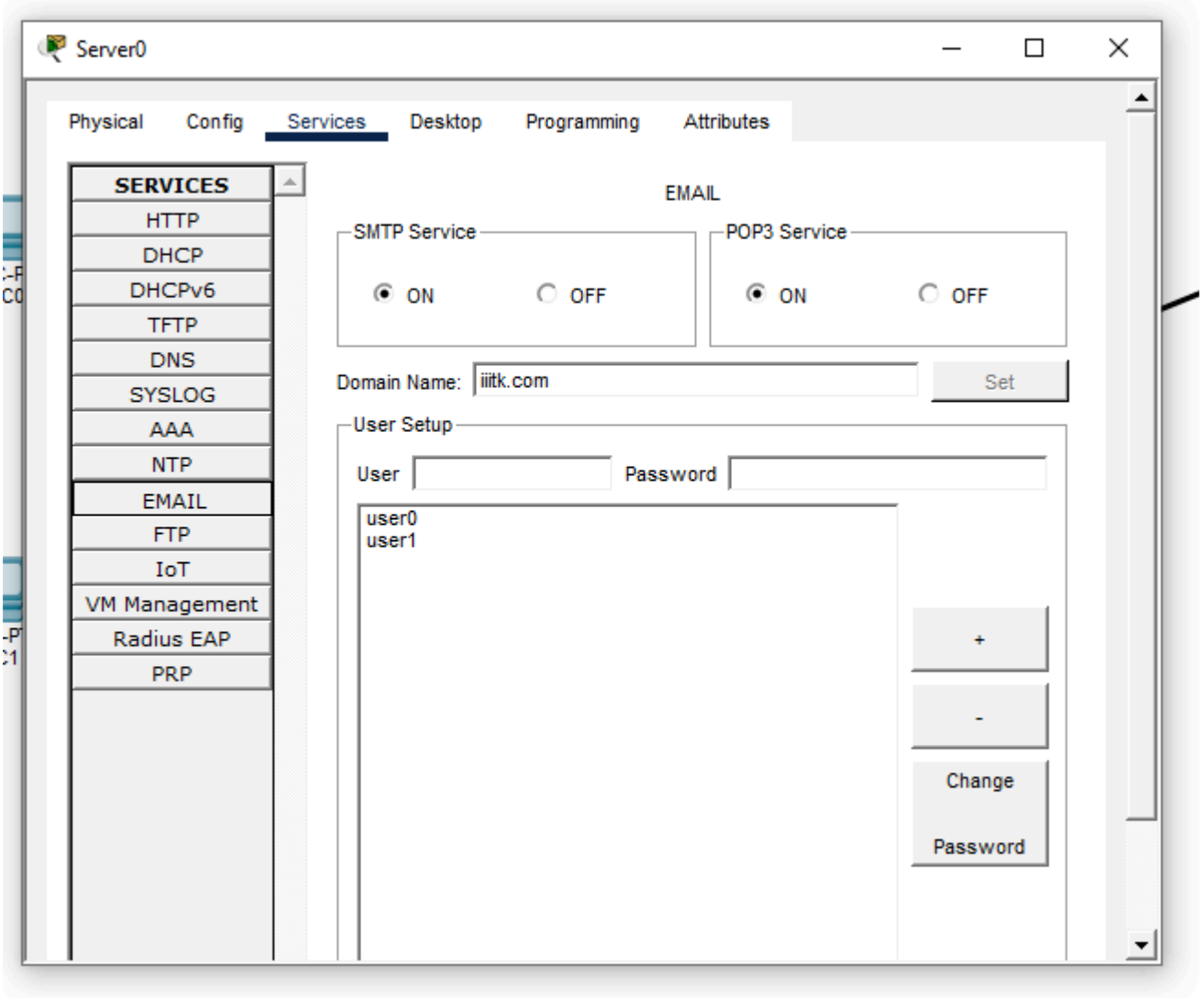
3. Router Configuration

CLI showing interface configuration.



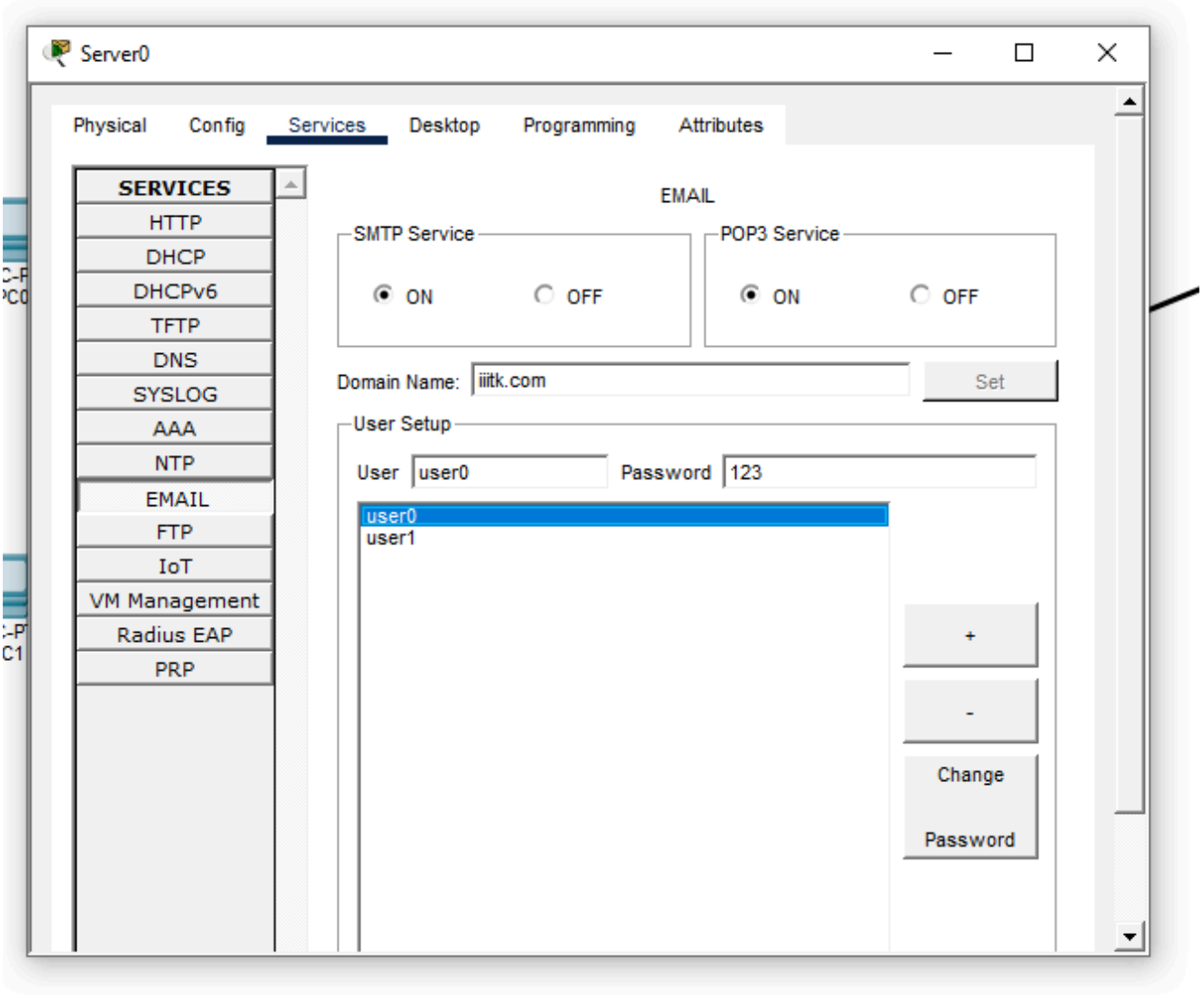
4. Email Server Configuration

SMTP and POP3 enabled.



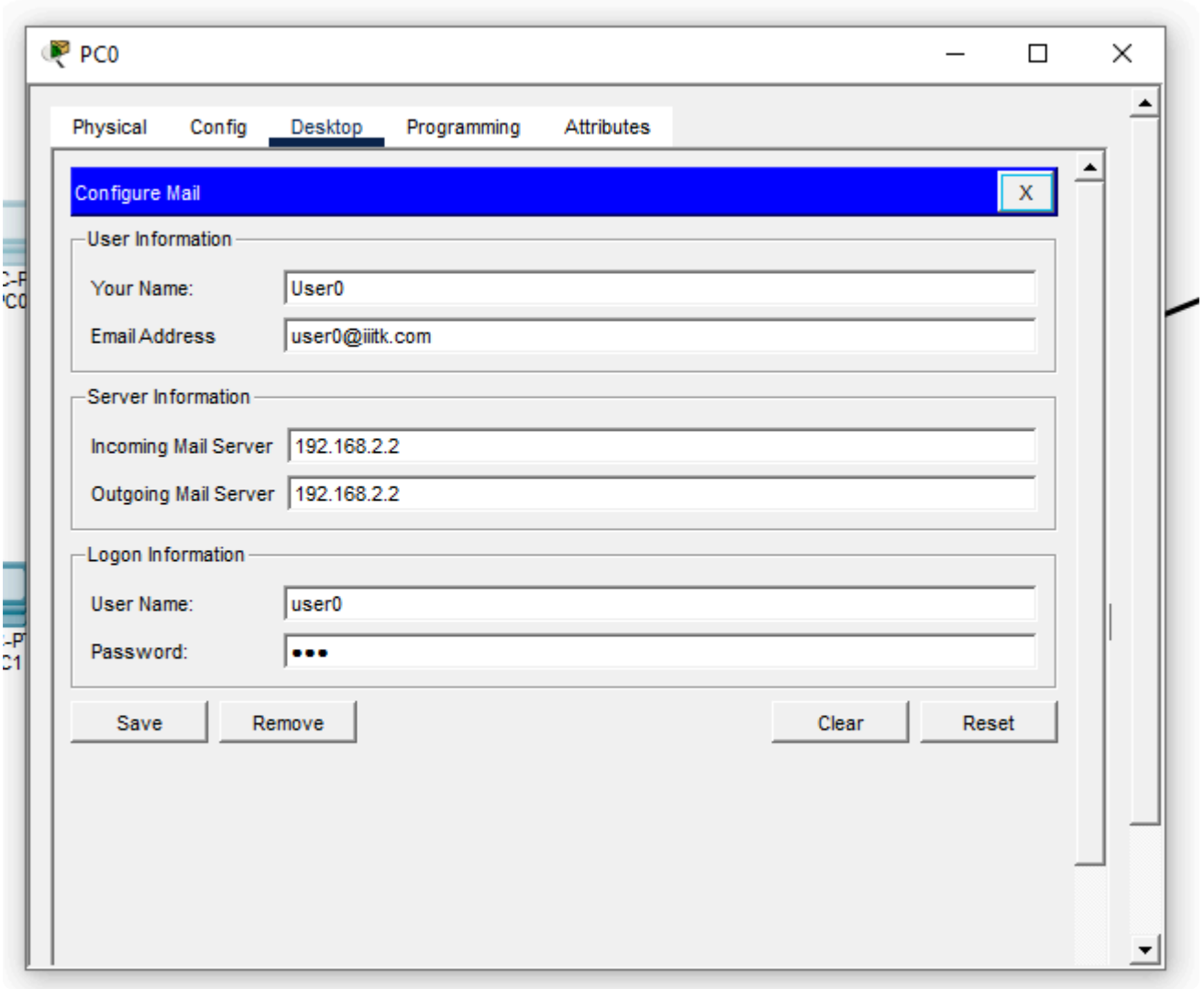
5. User Account Creation

user0 and user1 created on the email server.

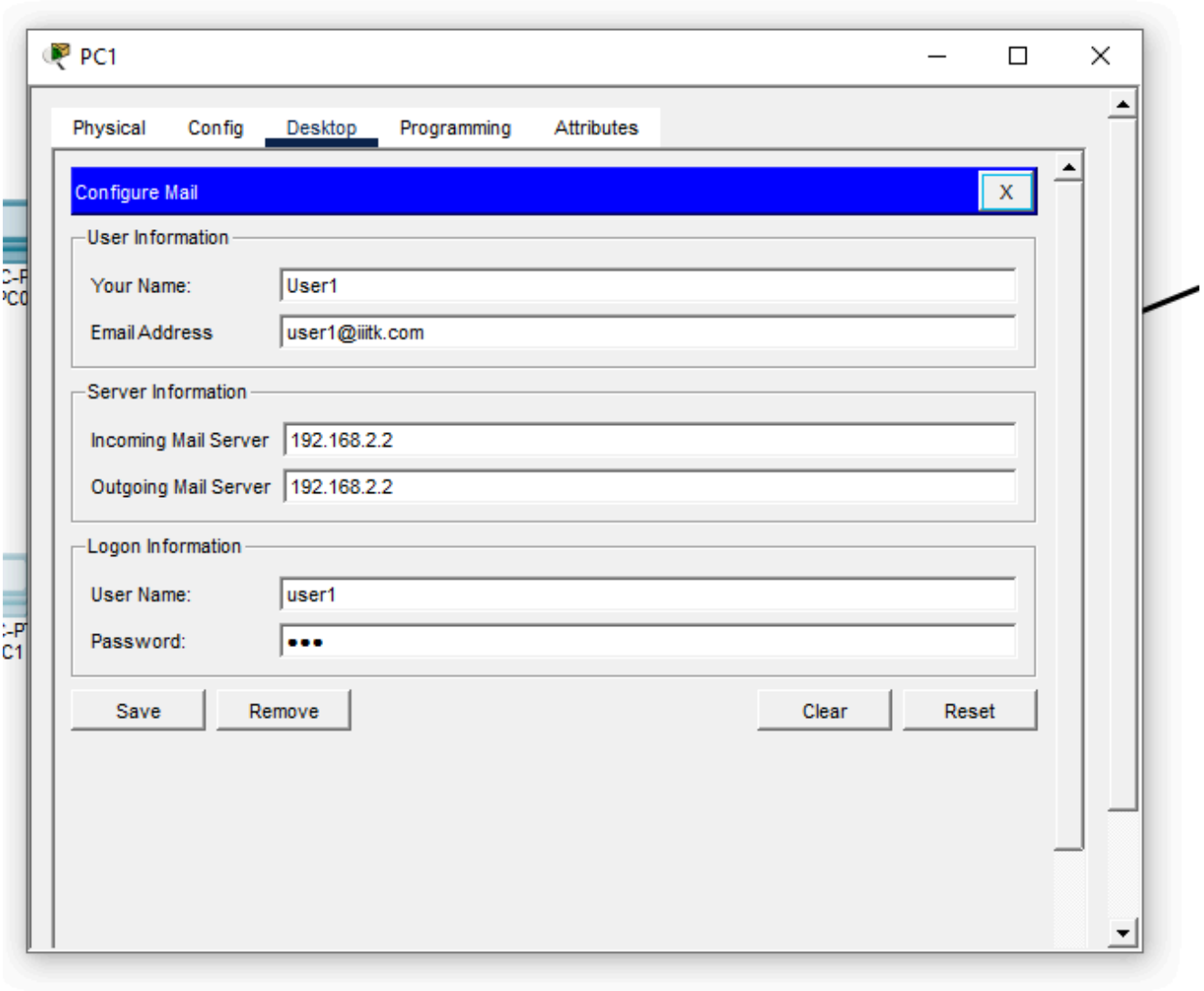


6. Email Client Configuration

PC0:

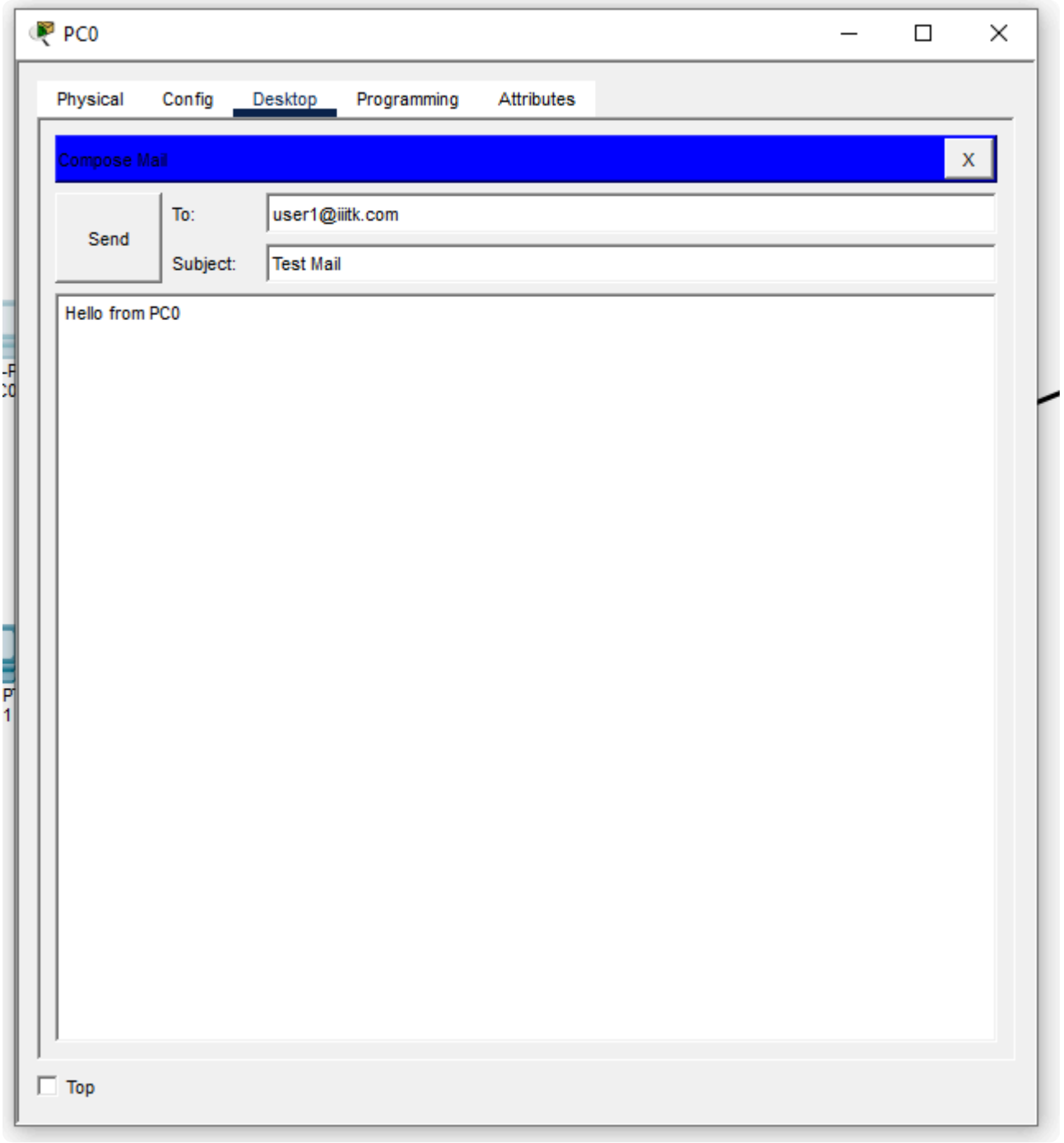

PC0
Physical
Config
Desktop
Programming
Attributes
Configure Mail
X
User Information
Your Name:
User0
Email Address
user0@iiitk.com
Server Information
Incoming Mail Server
192.168.2.2
Outgoing Mail Server
192.168.2.2
Logon Information
User Name:
user0
Password:
Save
Remove
Clear
Reset

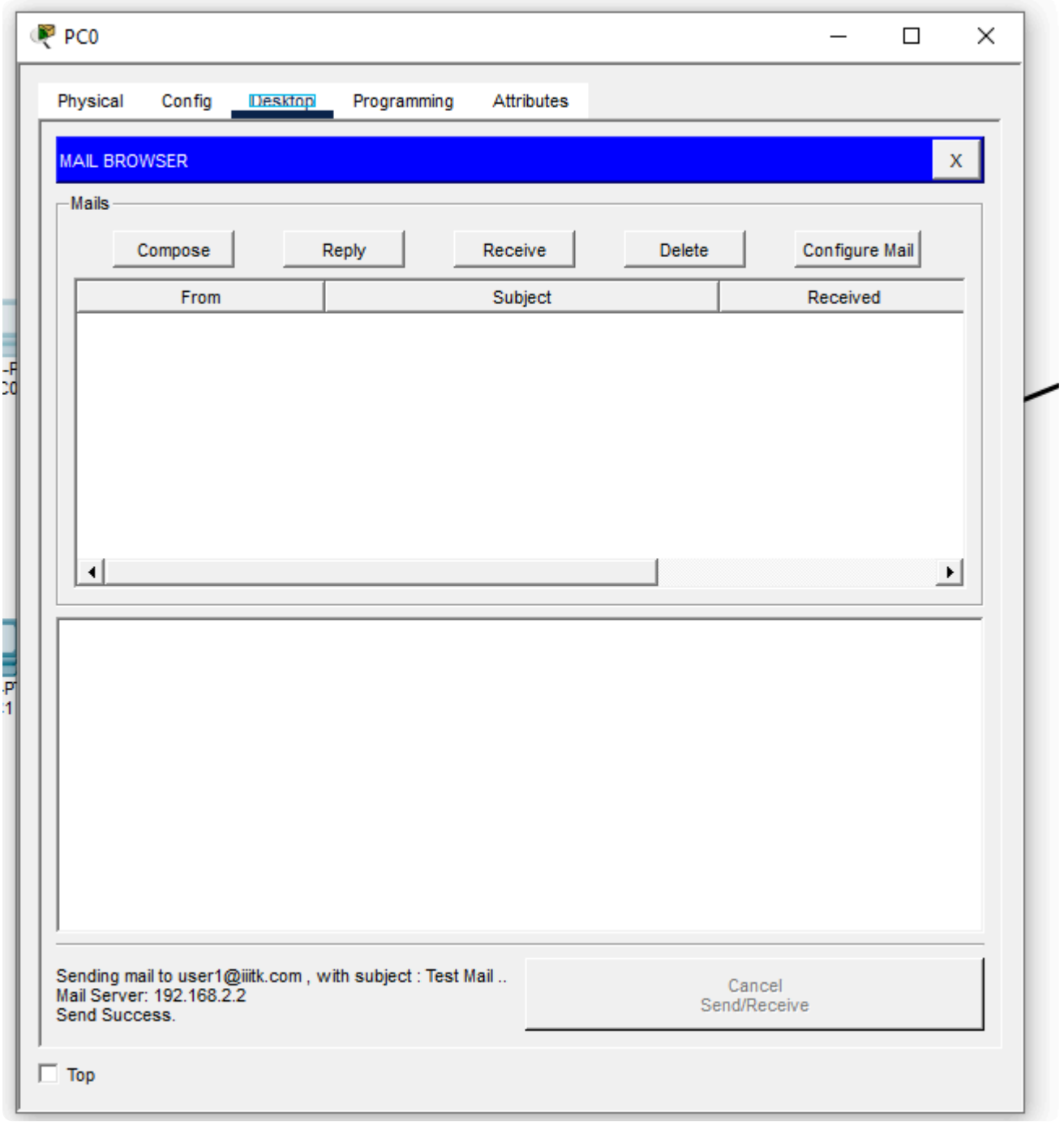
PC1:



7. Email Sent

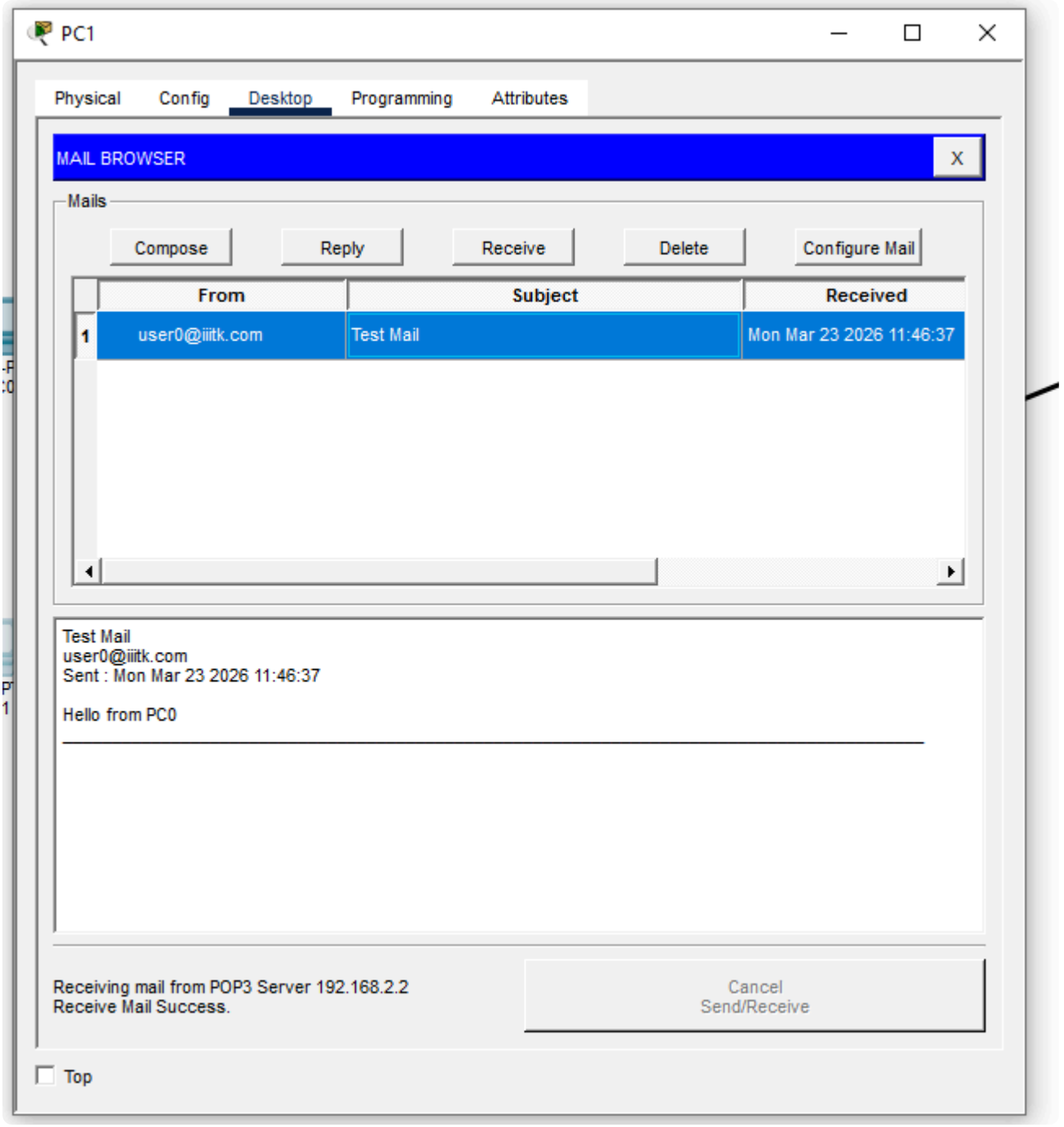
Message sent from PC0 to PC1.


PC0
Physical
Config
Desktop
Programming
Attributes
Compose Mail
X
Send
To:
user1@iiitk.com
Subject:
Test Mail
Hello from PC0
Top

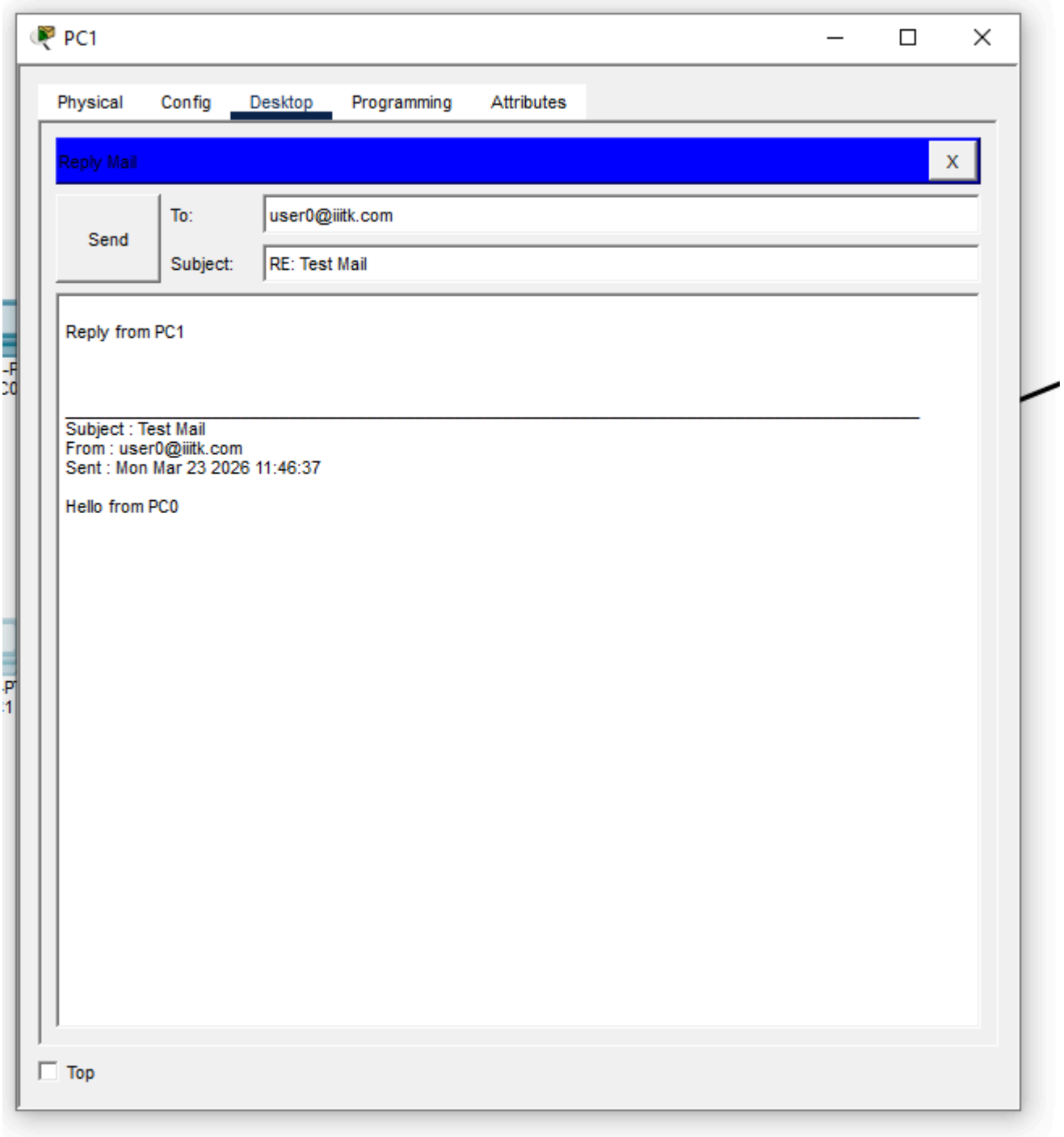


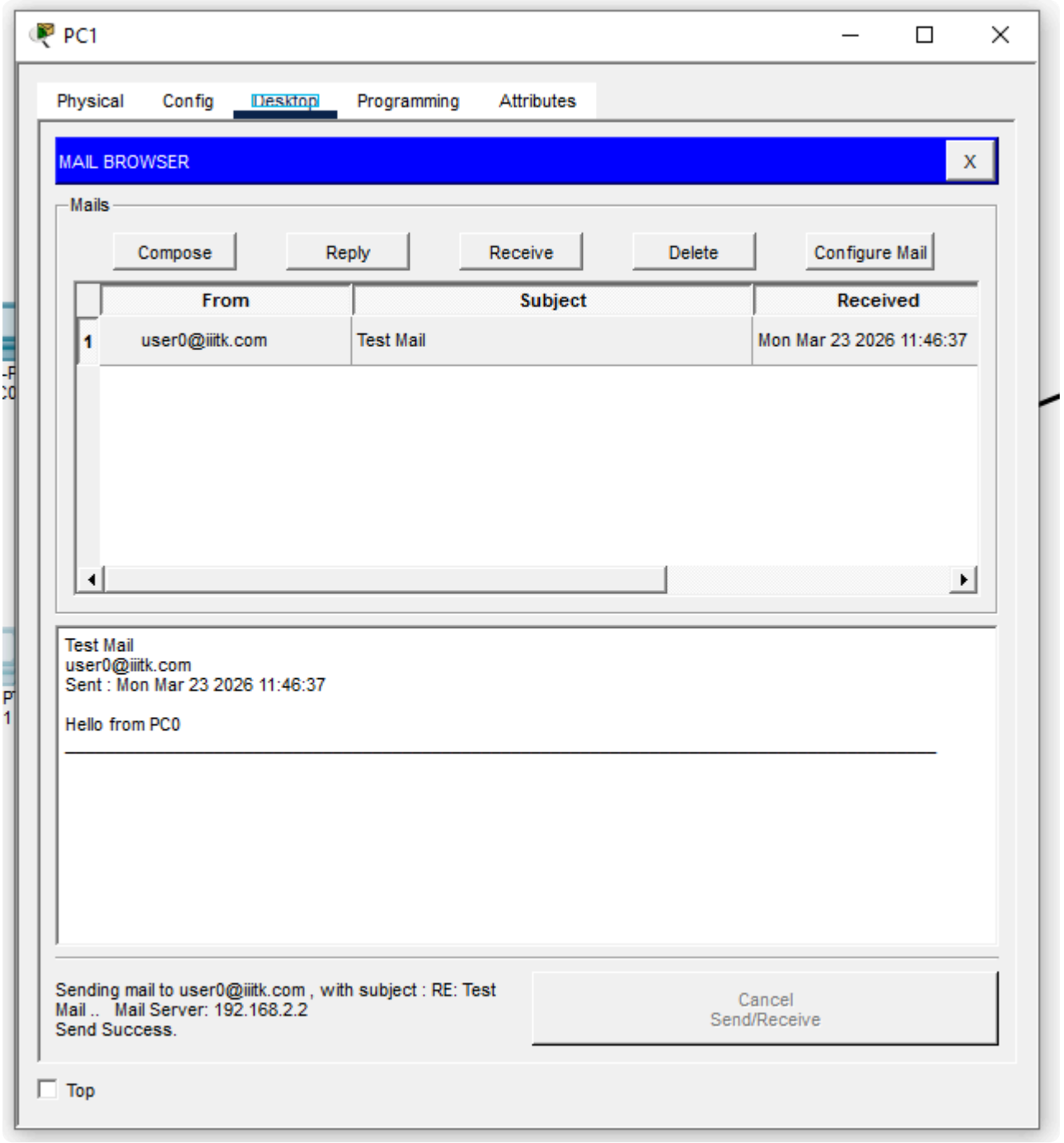
8. Email Received and Reply

Email received on PC1:



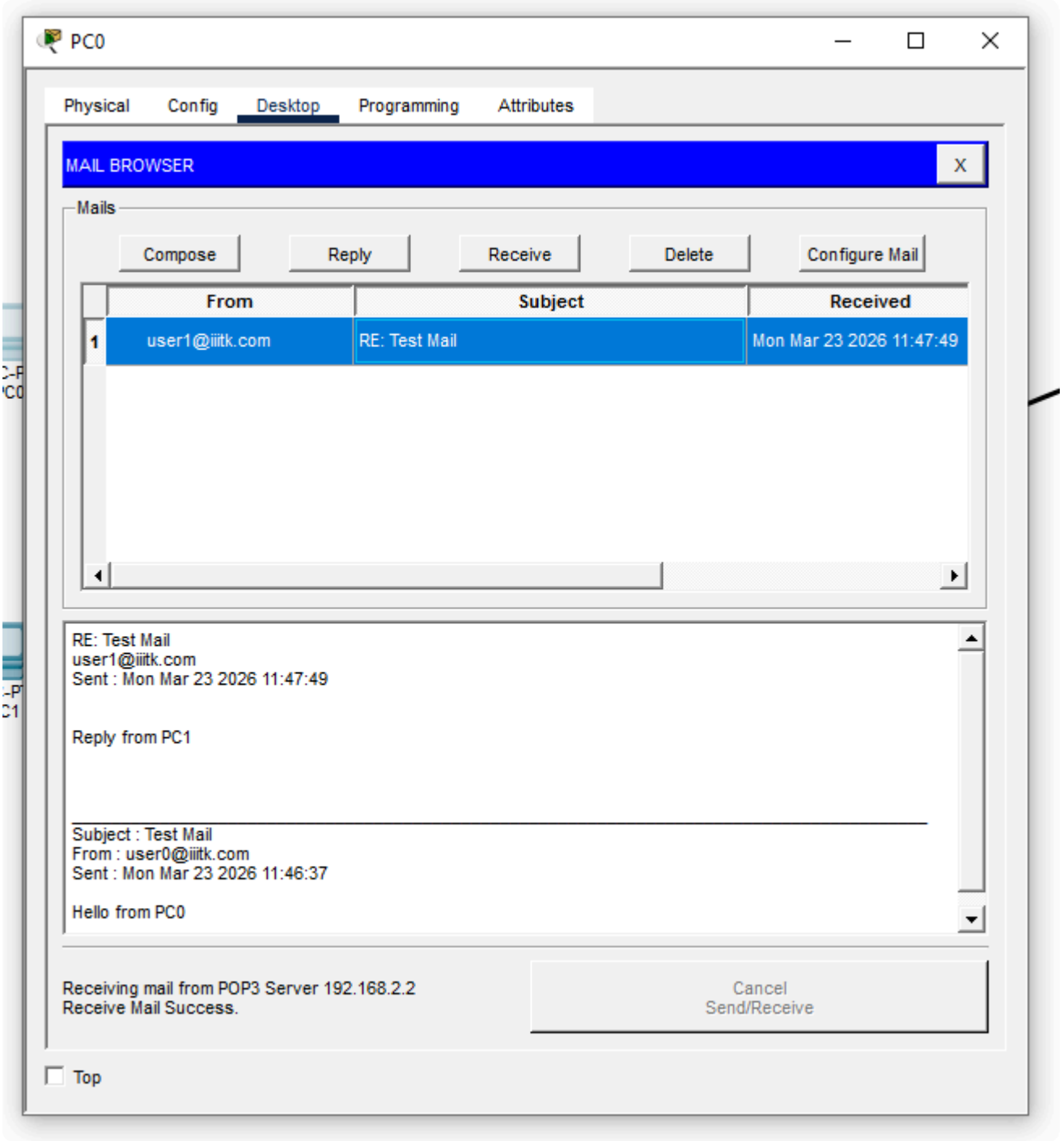
Reply sent from PC1 to PC0:


PC1
Physical
Config
Desktop
Programming
Attributes
Reply Mail
X
Send
To:
user0@iiitk.com
Subject:
RE: Test Mail
Reply from PC1
Subject : Test Mail
From : user0@iiitk.com
Sent : Mon Mar 23 2026 11:46:37
Hello from PC0
Top



Reply Received

Reply received at PC0:



Result

The network was successfully created in **Cisco Packet Tracer** with two different IP networks connected through a router. The **Email Server** was configured using **SMTP and POP3 services**, user accounts were created, and email communication between **PC0 and PC1** was successfully verified.